# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

## ASSESSMENT TOOL 2 – Industry Problem Solving Task

| | | | |
|---|---|---|---|
| **Course Code:** | CSA06 | **Course Title:** | Design and Analysis of Algorithms |
| **CO Assessed:** | **CO4:** Articulation (BL4, BL5) | **Assessment:** | Assessment Tool 2 – Industry Problem Solving Task |
| **Weightage:** | 20% of CIA | **Total Marks:** | 100 |
| **CO4 Statement:** | Solve optimization problems using dynamic programming and greedy strategies. | | |
| **Student Name:** | CH.POORNA TEJA | **Reg. No.:** | 192472149 |
| **Section / Batch:** | 2024 | **Date:** | 22-08-2026 |

**Instructions to Students**

- Answer the following question. Total Marks: 100.
- Carefully analyze the given questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues.
- Provide a thorough analysis with validated results and performance evaluation.

| Question Type | Focus Area | Questions |
|---|---|---|
| **Industry Problem Solving Task** | Focus on Solving optimization problems using dynamic programming and greedy strategies | Q1 |

## SECTION A – Industry Problem Solving Task

**Code of Conduct**

*I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

## RUBRICS FOR CALCULATION

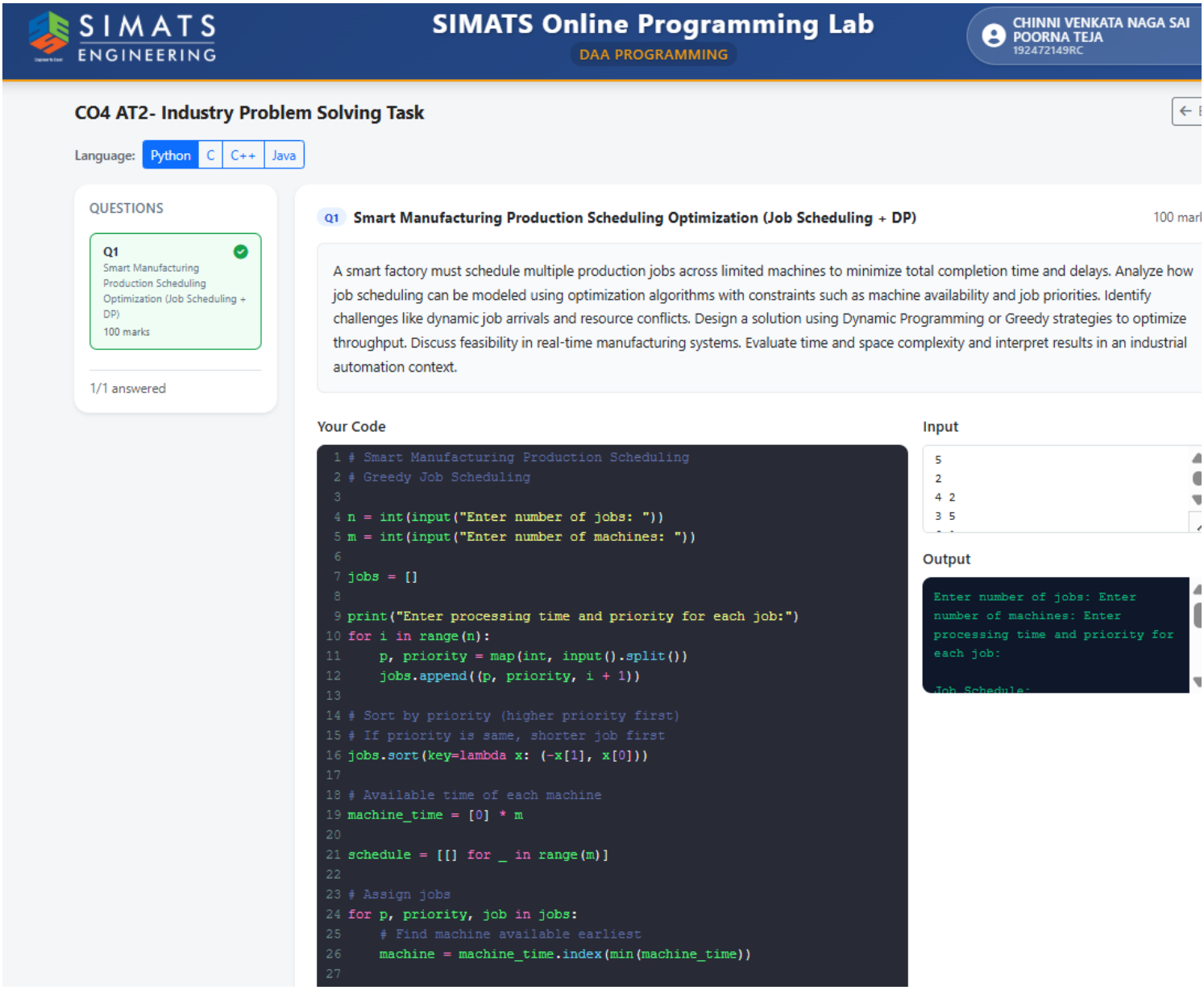
### Industry Problem Solving Task

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Industry Problem | 20% | Comprehensive understanding of real-world problem and constraints | Good understanding with minor gaps in requirements | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |
| Application of Engineering Concepts | 25% | Applies advanced and relevant concepts effectively to solve the problem | Applies appropriate concepts with minor errors | Limited application of concepts | Incorrect or no application of concepts |
| Solution Design | 25% | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps | Basic solution with limited feasibility | Unclear or impractical solution |
| Use of Tools | 15% | Effective use of modern tools, technologies, and real data | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools | Minimal or incorrect use of tools |
| Analysis | 10% | Thorough analysis with validated results and performance evaluation | Good analysis with reasonable validation | Basic analysis with limited validation | Poor or no analysis |
| Professionalism | 5% | Highly professional, well-documented, and clearly presented work | Good documentation and presentation | Basic documentation with some gaps | Poor documentation and presentation |

## Marks Evaluation:

| Criteria | Wt. | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Industry Problem | 20% | | | | |
| Application of Engineering Concepts | 25% | | | | |
| Solution Design | 25% | | | | |
| Use of Tools | 15% | | | | |
| Analysis | 10% | | | | |
| Professionalism | 5% | | | | |
| **Total Marks (100):** | | | | | |

| Faculty In-charge | Course Coordinator | Head of Department |
|---|---|---|
| Name: | Name: | Name: |
| Signature: _____ | Signature: _____ | Signature: _____ |
| Date: _____ | Date: _____ | Date: _____ |




**Algorithm: Smart Manufacturing Production Scheduling**

**Step 1:** Start.

**Step 2:** Read the number of jobs n and number of machines m.

**Step 3:** For each job, read:

- Processing time
- Priority
- Job number

**Step 4:** Store all jobs in a list.

**Step 5:** Sort the jobs:

- Higher priority jobs first.
- If priorities are equal, shorter processing time first.

**Step 6:** Initialize the available time of every machine to 0.

**Step 7:** For each job:

1. Find the machine with the **minimum available time**.

2. Assign the job to that machine.
3. Set the start time as the machine's available time.
4. Calculate:
   Finish Time = Start Time + Processing Time
5. Update the machine's available time.

**Step 8:** Display the schedule of all machines with job numbers, start times, and finish times.

**Step 9:** Calculate and display:

- Total completion time
- Overall completion time

**Step 10:** Stop.

**Complexity**

- **Sorting:** O(n log n)
- **Job assignment:** O(n × m)
- **Overall Time Complexity: O(n log n + n×m)**
- **Space Complexity: O(n + m)**

**Main idea:** Use a **Greedy strategy** by giving priority to important jobs and assigning each job to the machine that becomes available earliest.